1 — System Overview

> **Audience:** Everyone — no technical knowledge required.

What Does This System Do?

The **Pumping Station IoT Monitoring System** allows you to remotely watch and manage hundreds of water pump stations in real time — from any web browser, anywhere in the world.

Without this system, a field technician must physically drive to each pump station to check if it is running correctly. With this system, all that information is available instantly on a web dashboard.

Key Benefits at a Glance

Benefit	Detail
📡 **Real-time Monitoring**	Live electrical readings every 30 seconds from every station
🔔 **Automatic Alerts**	Email notifications the moment a pump overloads, stops, or voltage drops
📊 **Historical Data**	Full log of all readings — export to Excel/CSV for reporting
🌍 **Interactive Map**	See all stations on a map, color-coded by status
🔒 **Secure Access**	Role-based login (Admin, Operator, Viewer)
📈 **Scalable**	Supports 1 to 400+ stations with no infrastructure changes

---

System Components

```
┌──────────────────────────────────────────────────────────────┐
│                    PUMP STATION (on-site)                    │
│                                                              │
│  ┌──────────────┐     Modbus RTU      ┌───────────────────┐  │
│  │ PZEM-004T    │◄───────────────────►│ ESP32-S3 Board    │  │
│  │ Power Meter  │                     │ (Waveshare 4G)    │  │
│  └──────────────┘                     └─────────┬─────────┘  │
```

```
│                                          │ 4G LTE       │
└──────────────────────────────────────────┼─────────────┘
                                           │
                                           │
                                  ┌────────▼────────┐
                                  │ Google Firebase  │
                                  │ (Cloud Database) │
                                  └────────┬────────┘
                                           │
                                           │
                                  ┌────────▼────────┐
                                  │ Web Dashboard    │
                                  │ (Any Browser)    │
                                  └─────────────────┘
```

Component Descriptions

1. PZEM-004T AC Power Meter (On-site Sensor)

- Clamps onto the pump's electrical cable inside the cabinet
- Measures: **Voltage (V), Current (A), Power (W), Energy (kWh), Frequency (Hz), Power Factor**
- Communicates with the main board via a short cable (Modbus protocol)
- Requires no external power — draws power from the ESP32 board's 5V rail

2. Waveshare ESP32-S3 A7670E 4G Board (The "Brain")

- A small computer installed inside the electrical cabinet
- Reads sensor data every 5 seconds, averages it, then uploads every 30 seconds
- Connects to the internet via a **4G SIM card** (no Wi-Fi needed)
- Has an onboard **GPS** — automatically determines and reports its own location
- Has a **status LED** to indicate operational state at a glance

3. Google Firebase Cloud (Data Storage & Logic)

- Stores all live and historical readings securely in the cloud
- Triggers automatic alert emails when thresholds are exceeded
- Free tier is sufficient for the pilot; cost scales linearly for 400+ stations

4. Web Dashboard (Your Control Panel)

- Accessible from any modern web browser (Chrome, Firefox, Safari, Edge)
- No app installation required
- Features: live metrics, map view, alert history, data export, station management

---

What Data Is Collected?

Every 30 seconds, each station reports:

Metric	Unit	Purpose
Voltage	V	Detect over/under voltage (grid faults)
Current	A	Detect pump overload or pump failure
Active Power	W	Monitor energy consumption
Energy	kWh	Billing and efficiency tracking
Frequency	Hz	Detect grid instability
Power Factor	—	Detect electrical efficiency issues
GPS Location	lat/lon	Automatic station mapping
Online Status	—	Is the device reachable?

Alert System

The system sends email alerts automatically for:

- 🔴 **HIGH CURRENT** — pump drawing too much power (overload / blockage)
- 🔵 **LOW CURRENT** — pump stopped or tripped (dry run / power cut)
- 🟡 **HIGH VOLTAGE** — grid overvoltage (risk to pump motor)
- 🟡 **LOW VOLTAGE** — grid undervoltage (pump running inefficiently)
- ⚫ **DEVICE OFFLINE** — communication lost for more than 5 minutes

All thresholds are configurable per station from the dashboard.

User Roles

Role	Can Do
Admin	Everything — manage users, stations, settings
Operator	View all stations, acknowledge alerts, export data
Viewer	Read-only access to dashboard

Next: [Hardware Setup & Wiring →]